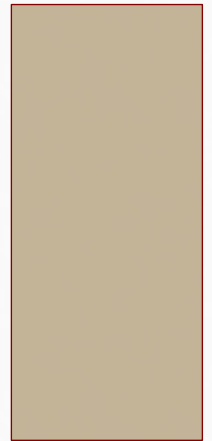


REPRESENTING A 3D ENVIRONMENT

ADVANCED GAME DESIGN & DEVELOPMENT

DANNY JUGAN



3D ENVIRONMENTS

- There are two parts of the **Transform** component that we need to look at.
- **Position** (Vector3)
 - Points in space (x, y, z)
- **Rotation** (Quaternions / Euler Angles)
 - Euler Angles are representations of rotations on the three axes (x, y, z)
 - Don't mess with Quaternions unless you know what you're doing

MOVING OBJECTS

- Let's move some objects around in Unity using various `Vector3` positions and Euler rotations
 - Creating Vectors and Eulers in code
 - Moving objects over time

LERP AND SLERP

- **Lerp** – Linearly interpolates between two vectors
- **Slerp** – Spherically interpolates between two vectors.
 - `static Vector3 Lerp(Vector3 from, Vector3 to, float t);`
 - Interpolates between from and to by the fraction t.

EXERCISE

- How would you get the yellow arrow to move back and forth along the aim arc at the top?

